BEING AT-WORST-NODAL OF RELATIVE DIMENSION 1 IS FPQC LOCAL ON THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the field case of the proof invokes \label{lemma-node-field-extension} (source line 5090), a published lemma of the same chapter whose own proof delegates its central ring-isomorphism step to the reader (source lines 4646-4652: 'We encourage the reader to prove this for themselves ... We omit the details.'); that lemma was therefore not carded separately, but it is used here as published input per the citation policy; the remaining steps of the present proof are complete.

Editorial note. The label lemma-nodal-family-fpqc-local-target names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Base change of nodal families. Same chapter, Lemma \label{lemma-base-change-nodal-family}. Proof lines 5078-5079: the direction (1) implies (2) follows from the base change lemma for nodal families.
- Descent of the structural properties along fpqc coverings. Descent, Lemmas \label{descent-lemma-descending-property-locally-finite-presentation} and \label{descent-lemma-descending-property-flat}; Morphisms, Lemma \label{morphisms-lemma-dimension-fibre-after-base-change}. Proof lines 5081-5087: being locally of finite presentation, flat, or of relative dimension 1 is fpqc local on the target, which reduces the remaining direction to the case of a field.
- Nodes under field extension. Same chapter, Lemma \label{lemma-node-field-extension}. Proof lines 5090-5093: over a field the statement follows from this lemma, together with the fpqc-locality of smoothness (Descent, Lemma \label{descent-lemma-descending-property-smooth}).

Source reference. Original lemma 02L2: sources/stacks-chow/tags.txt, source lines 3393–3393. Prerequisite pointer invoked at proof line(s) 5086; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02KY: sources/stacks-chow/tags.txt, source lines 3389–3389. Prerequisite pointer invoked at proof line(s) 5085; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02VL: sources/stacks-chow/tags.txt, source lines 3772–3772. Prerequisite pointer invoked at proof line(s) 5092; it names a standard

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C5B: sources/stacks-curves/pinned-original-source.tex, source lines 4932–4932. Prerequisite pointer invoked at proof line(s) 5080; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C56: sources/stacks-curves/pinned-original-source.tex, source lines 4605–4605. Prerequisite pointer invoked at proof line(s) 5090; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02FY: sources/stacks-chow/tags.txt, source lines 3209–3209. Prerequisite pointer invoked at proof line(s) 5087; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\{U_i \to S\}$ be an fpqc covering. The following are equivalent*

(1) *f is at-worst-nodal of relative dimension 1,*
(2) *each $X \times_S U_i \to U_i$ is at-worst-nodal of relative dimension 1.*

In other words, being at-worst-nodal of relative dimension 1 is fpqc local on the target.

Proof. One direction we have seen in Lemma 0C5B. For the other direction, observe that being locally of finite presentation, flat, or to have relative dimension 1 is fpqc local on the target (Descent, Lemmas 02KY, 02L2, and Morphisms, Lemma 02FY). Taking fibres we reduce to the case where S is the spectrum of a field. In this case the result follows from Lemma 0C56 (and the fact that being smooth is fpqc local on the target by Descent, Lemma 02VL). □